

Name_____

Statistics 1040: Introduction to Statistics

Section 7

Test 1 – February 15, 2001

75 minutes; closed book, closed notes, calculators and provided formula sheet allowed. 15 questions, 7 points each (max score is 105). For numeric questions, show your work and circle your answers. For essay questions, answer the questions in complete sentences. Good luck!

Use the following information for questions 1-3.

In a 4-year study of a random sample of 8,474 people with normal blood pressure, their anger was measured using a specially designed test that measures how prone a person is to sudden anger. At the end of the study, physicians (who did not know the anger test results) determined whether the people had heart disease or not, with the following results. Of the 3110 people classified as “low anger,” 1.7% had heart disease. Among the 4731 “moderate anger” subjects, 2.3% had heart disease, and 4.3% of the 633 “high anger” subjects had heart disease.

1. Is this an example of a randomized controlled experiment or of an observational study? Explain.

2. One critic of the study suggests that gender might be a confounding factor. Clearly explain what this means, and describe how the researchers might refute this criticism.

3. Do the above results prove that anger causes heart disease? Why or why not?

4. Armadillos are even more fond of sleep than college students and statistics professors. The following are the recorded number of hours slept per day for five different nine-banded armadillos:

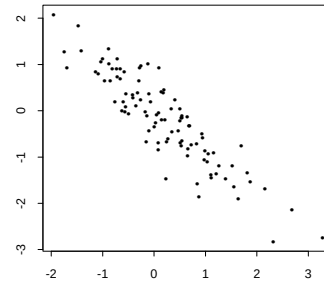
17, 18, 20, 16, 5

Find the average and median of this dataset. (Show your work!) Which is a more appropriate measure of center for this data? Why?

5. Air Force flight instructors once observed that very good landings were usually followed by landings that were not so good, while very poor landings were usually followed by somewhat better landings. The instructors reasoned that this pattern occurred because they had praised the good landings and harshly criticized the poor ones. Thus they concluded that praise is detrimental to learning, and severe criticism is beneficial. Identify an alternative, statistical, explanation for the instructors' observations.

6. For the following scatterplot, which of the following values is the most reasonable estimate of r for the data? Why?

-1.5 -0.9 -0.6 0 0.6 0.9 1.5



For questions 7 and 8 use the following data.

The British government conducts regular surveys of household spending. This table shows average weekly household spending in British pounds on tobacco products and alcoholic beverages for 10 regions of the United Kingdom in 1981. The correlation coefficient (r) between these 10 pairs is 0.78.

Region	Alcohol	Tobacco
North	£6.47	£4.03
Yorkshire	6.13	3.76
Northeast	6.19	3.77
East Midlands	4.89	3.34
West Midlands	5.63	3.47
East Anglia	4.52	2.92
Southeast	5.89	3.20
Southwest	4.79	2.71
Wales	5.27	3.53
Scotland	6.08	4.51
Average	5.59	3.52
SD	0.68	0.53

7. If you were to calculate r for weekly household alcohol and tobacco expenditures for all the individual households in these 10 regions, would you expect it to be larger than 0.78, about 0.78, or less than 0.78? Explain.
8. In 1981, one pound was worth about 1.90 U.S. dollars. If we converted the 20 above expenditures into U.S. dollars, how would this change the average and SD of tobacco spending, and the correlation r ? Support your answers with calculations or short explanations.

For questions 9 to 12, use the following dataset:

Over the past century, melanoma (skin cancer) rates have been steadily increasing. A plot of year vs. Connecticut melanoma cases per 100,000 inhabitants for the years 1936–1972 is roughly elliptical, with the following summary statistics.

$$\begin{array}{ll} \text{Average year} = 1954 & \text{SD} = 10.8 \\ \text{Average melanoma rate} = 2.8 & \text{SD} = 1.2 \quad r = 0.96 \end{array}$$

9. Assuming the histogram of melanoma rates is well fit by the normal curve, about what percentage of years had a melanoma rate of greater than 4 (cases per 100,000)?

10. Find the equation for predicting melanoma rate from year, and use it to estimate the rate for the year 2000.

11. Suppose we estimated melanoma rate for the year 1960. Would we expect this prediction to be much more accurate, much less accurate, or about as accurate as the prediction in question 10? Why? (*Note: You do not need to actually calculate this value.*)

12. Calculate the likely size of the residual from the 1960 estimate.

For questions 13 and 14, use the following data set on ages of cars in the U.S. in 1994.

Age (Years)		Percentage of Cars		
at least	but no more than			
0	6	36		
6	9	21		
9	12	18		
12	18	24		

13. Draw the histogram on the density scale. Label all axes and tick marks.

14. Using the histogram, find the percentage of cars in 1994 which were less than 8 years old.

15. A doctor wishes to conduct an experiment on the effects of small regular doses of aspirin on heart attack rates in women. She has 50 female patients willing to participate in her experiment. She selects 25 to form her treatment group and leaves the other 25 as a control group, trying to balance the two groups for factors such as excess weight. The treatment group will take two aspirin every day for five years, while the control group takes nothing. At the end of the study, the doctor will compare the number of heart attacks in the two groups over the five-year period. Identify at least two ways in which this study could be improved while using only the same 50 patients, and explain why those are good ideas.